Stanley Nelavala

Platform and Site Reliability Engineer

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Huntington, IN | Open to relocate anywhere

Summary

Backend engineer focused on reliability. Public projects define SLIs with 28-day error budgets and promtool-tested burn-rate alerts, run scripted failure labs with invariant checks, roll back by image digest, and size capacity with a Little's law model (local benchmark). At work, cut dev and test server runtime from 168 to about 53 hours a week and standardized upload malware scanning across 9 modules. M.S. in Computer Science (4.0 GPA). AWS Certified Solutions Architect, Associate.

Experience

Software Development Engineer Nov 2025 – Present
Mass Technologies, team of 10 (RxApps360 hospital platform, pre-launch; numbers from dev and test) *Huntington, IN*

- Automated weekday start and stop of dev and acceptance servers and databases (AWS ECS, RDS): they run about 53 hours a week instead of 168.
- Standardized malware scanning of file uploads (AWS GuardDuty) across 9 app modules.
- Set up separate iOS and Android release pipelines in GitHub Actions with automatic TestFlight submission.
- Designed, built and integrated into every platform module the signed service-to-service API (M2M): each call proves which service and tenant sent it. After a security audit, hardened it with a Redis rate limiter and replay protection.
- Replaced offset paging with keyset paging on the catalog device list: on a 328,000-device dev database, the query for row 150,000 fell from 285 ms to 0.20 ms. Search pages never skip or repeat rows, tested with rows inserted, deleted and tied between pages.

Adjunct Faculty (part-time) Jan 2026 – Aug 2026
The College of St. Scholastica *Duluth, MN*

- Taught Programming II (Java) and Machine Learning for Data Science.

Graduate Research Assistant and Teaching Assistant Aug 2023 – Aug 2025
University of Minnesota Duluth *Duluth, MN*

- Built a pipeline (Python, Node.js, MySQL, FFmpeg, PaddleOCR) that collected and analyzed 1M+ TikTok videos from 300k+ authors for thesis research on misinformation, then grouped themes with BERTopic; presented at the UMD CS graduate poster session. Led labs and office hours for Python, NLP and Parallel Programming.

Projects (public code, tests)

Payments Ledger Service | *Java 25, Spring Boot, PostgreSQL, Flyway, Kafka API, Prometheus, Grafana, k6* GitHub

- Three user-facing SLIs with 28-day error budgets and six multi-window burn-rate alerts tested with promtool; a scripted failure lab (Kafka down, Postgres restart, relay crash, poison event, latency injection) with balance checks after each.
- Rolled forward and back by image digest across an expand-and-contract Flyway migration; a k6 SLO gate fails a forced slow run; dead-letter topic with once-only redrive.

Inference Gateway | *Go, gRPC, ONNX Runtime, PyTorch, OpenTelemetry, Prometheus, Helm* GitHub

- Canary controller with a weighted proxy and an error and p95 gate that rolled a faulty version back on its own; latency SLO alerts tested with promtool; Helm chart tested on kind.
- One-factor experiments and pprof ruled out GC and tokenizer cost; a Little's law model sizes a 10x launch peak at 2 to 3 replicas (local benchmark).

Webhook and Connector Gateway | *Rust, axum, tokio, PostgreSQL, SQLite, sqlx, OpenTelemetry, k6* GitHub

- Global backpressure (503 with Retry-After) and a kill -9 failure lab where 500 of 500 events reached the downstream once after redrive; delivery SLO alerts tested with promtool.

Skills

Cloud: AWS (ECS, RDS, S3, SQS, Secrets Manager, GuardDuty), Terraform, CloudFormation (lint), Docker, Kubernetes (kind), Helm
Delivery: GitHub Actions (reusable workflows, OIDC), canary rollout and rollback, rollback by image digest, expand-and-contract migrations
Observability: OpenTelemetry, Jaeger, Prometheus, Grafana, SLIs, SLOs and burn-rate alerts (promtool), blackbox probes, pprof
Reliability and security: failure labs and runbooks, Toxiproxy, k6, Locust, capacity models, Trivy, gitleaks, CycloneDX SBOM
Languages: Python, Go, Rust, Java, Bash, SQL, TypeScript

Education and Certification

M.S. Computer Science, University of Minnesota Duluth, 2023 – 2025, GPA 4.0, Graduate Summer Research Fellowship 2024 and 2025
B.Tech Computer Science and Engineering, IIIT D&M Kancheepuram, 2019 – 2023
AWS Certified Solutions Architect, Associate, Feb 2025 – Feb 2028 (verifiable on Credly)